

CSE208: Problem on Intractability

ID-1:

ID-2:

Problem Definition

A graph $G = (V, E)$ and an integer $k \geq 3$.
K Coloring Problem: Can the vertices of the graph G be colored using at most k colors such that no two adjacent vertices share the same color?
This means finding a function $c : V \rightarrow \{1, 2, \dots, k\}$ such that for every edge $(u, v) \in E$, we have $c(u) \neq c(v)$.
The **chromatic number** of a graph G , is the minimum number of colors needed to color the vertices of the graph such that no two adjacent vertices share the same color.

Task 1: Find the chromatic numbers for the bipartite and complete graphs.

First, we need to prove that K-Coloring problem is in NP. For a given graph and its coloring, we need to find a way to verify in polynomial time.

Task 2: Prove that K-Coloring problem is in NP.

Now, we will show that K-Coloring problem is NP-hard. We will choose a renowned NP-complete problem (3-coloring problem), reduce an arbitrary instance of 3-coloring problem to an "equivalent" instance of k-coloring problem. By 'equivalent', we mean that the original instance of 3-coloring problem is satisfiable if and only if the transformed instance of k-coloring Problem is satisfiable

Task 3: Describe how to transform an instance of 3-coloring problem into an 'equivalent' instance of k-coloring Problem. You also have to show this transformation runs in polynomial time.

Task 4: Prove that the transformation you described in Task 3 is indeed a polynomial-time reduction from 3-coloring to k-coloring Problem. In other words, show that,

- If the original 3-coloring instance is satisfiable, then the transformed k-coloring instance is also satisfiable.
- If the original 3-coloring instance is not satisfiable, then the transformed k-coloring instance is also not satisfiable.

OR equivalently, If the transformed k-coloring instance is satisfiable, then the original 3-coloring instance is also satisfiable.